Gokul Krishna Balaji

+91 98407 86808 gokulkrishna1686@gmail.com linkedin.com/in/gokul-krishna-balaji
github.com/gokulkrishna1686

Education

Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram Chennai
Bachelor of Technology in Computer Science and Engineering (AI); ***CGPA: 9.15/10.0*** *Expected 2028*

- **Relevant Coursework:** Data Structures, Algorithms, Database Systems, Artificial Intelligence, Computer Organization and Architecture, Problem Solving and Programming

Technical Skills

Languages: Python, C, C++, JavaScript, SQL, HTML, CSS
Web: React.js, Next.js, Node.js, Express.js, FastAPI, Flask, Tailwind CSS, MongoDB, MySQL, Convex
Machine Learning Frameworks: PyTorch, Scikit-learn, XGBoost, OpenCV, NumPy, Pandas
Tools: Git, GitHub, Linux, Docker, Postman, Vercel, Render

Experience

Reelcat Remote
Software Development Engineer (SDE) Intern *Dec 2025 – Jan 2026*

- Built an AI-powered video editing platform using **React**, **TypeScript**, and **Convex**.
- Developed scalable video rendering pipelines using **AWS Remotion Lambda**.
- Integrated **OpenAI** for transcription and smart captions, reducing manual editing by **70%**.
- Implemented **Razorpay** payments and **Resend** transactional emails.

Projects

Redpen AI | *Next.js, FastAPI, Supabase, Gemini* Jan 2026

- Built an **AI-powered grading system** for handwritten subjective exams using a multimodal LLM with rubric-constrained evaluation and structured JSON outputs with reasoning.
- Designed professor and student interfaces supporting one-time rubric configuration, automated batch processing of answer scripts, and analytics dashboards for performance insights.
- Secured **1st Prize among 80+ teams** at E-Summit '26 Ideathon.

ReFile | *FastAPI, OpenAI API, Docker, Linux* Oct 2025

- Architected a **SaaS file transformation engine** that converts natural language prompts into executable Linux CLI commands using OpenAI, enabling dynamic media and file manipulation.
- Executed AI-generated commands (e.g., ffmpeg pipelines) inside **isolated Docker containers** to ensure secure, sandboxed, and scalable processing.
- Awarded **2nd Place** at the VØID:V1 National Hackathon for innovation and system design.

VectorNote (RAG System) | *Flask, LangChain, Chroma, Gemini API* Aug 2025

- Engineered an end-to-end **Retrieval-Augmented Generation (RAG) pipeline** for multi-PDF querying using recursive chunking (400/50 overlap), Gemini embeddings, and persistent Chroma vector storage.
- Implemented **incremental indexing** with metadata-based document tracking, top-k (k=10) similarity retrieval, and conversational memory integration for context-aware multi-turn reasoning.
- Optimized backend performance via async execution with ThreadPoolExecutor and **state-aware re-indexing** to eliminate redundant embedding operations.

Positions of Responsibility

Google Developer Student Club (GDSC) Chennai, India
Coordinator *Aug 2025 – Present*

- Built a **RAG-based chatbot** to answer event and resource queries for **500+ students**.
- Conducted workshops on AI APIs, automation tools, and full-stack development for **150+ students**.